

Anushka Pradeep Chauhan

Robotics Engineer - Perception & Manipulation

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Summary

Robotics grad student focused on perception-to-actuation pipelines for manipulation - RGB-D fusion, YOLO-based detection, SLAM - built in ROS2, tested in simulation, and deployed on real hardware. Co-authored a paper on autonomous robotic sorting, published at IEEE ICNEWS 2024.

Education

Northeastern University

Dec 2026

Master of Science in Robotics, Electrical and Computer Engineering

Boston, MA

- Relevant Courses: Mobile Robotics, Robot Sensing & Navigation, Robotics Mechanics & Control, Assistive Robotics, Human-Robot Interaction, Robot Science & Systems, Legged Robotics, Pattern Recognition & Computer Vision

University of Mumbai

May 2024

Bachelor of Engineering in Electronics and Telecommunication

Mumbai, India

Technical Skills

Programming: Python, C++, MATLAB, Linux, Git, Docker

ROS & Robotics: ROS, ROS2, MoveIt2, TF2, SLAM, Navigation, Motion Planning, Robot Manipulation

Perception: OpenCV, YOLOv8, Vision Transformers, Depth Estimation, LiDAR + RGB-D Sensor Fusion

Simulation & Tools: Gazebo, NVIDIA Isaac Sim, USD Scene Graph, Real-to-Sim Validation

Hardware: RealSense D405, Kinect RGB-D, Raspberry Pi, Embedded Linux Systems

Projects

Autonomous Drink Pouring - Stretch RE3 Mobile Manipulator, Northeastern University

Feb 2025 - Apr 2025

- Hit **90% (8/10) autonomous grasp-and-pour success** by fusing ArUco tracking on a gripper-mounted RealSense D405 with a YOLOv5n detector (1.9M params, 25-30ms/frame) on the head-mounted D435
- TF2 transforms and camera-intrinsic reprojection turned raw detections into stable base_link coordinates, and a rolling frame average kept the motion planner working off jitter-free targets instead of noisy single-frame reads
- Built a lift-arm-wrist grasp controller with ± 0.7 rad wrist-roll correction, tested first in a custom Gazebo world before touching hardware, so pours ran on **embedded Linux** with zero spillage across all 10 real trials

Industrial Picker Arm (Published, IEEE ICNEWS 2024), University of Mumbai

Sep 2023 - Mar 2024

- Trained **YOLOv8** on a self-annotated 2,000-image dataset (3 package classes) and integrated it into a 6-servo, 360° arm we built from scratch (60kg-torque shoulder, dual 25kg elbow servos, suction end-effector) - **classification errors dropped 35%**
- Combined RGB detections with Kinect depth for x/y/z centroid extraction, driving pick-and-place motion through a **Linux (Raspbian)**-based Raspberry Pi 4 + PCA9685 PWM controller; automating trajectory generation cut **handling errors 20%**
- Cut **end-to-end response time 30%** by simplifying the ROS messaging between vision and motion nodes, which mattered for segregating packages from a cluttered bin in real time

riSAM: Robust Incremental Smoothing & Mapping, Northeastern University

Sep 2024 - Dec 2024

- Implemented pose-graph SLAM on **Linux (Ubuntu)** with Graduated Non-Convexity and a Scale-Invariant Graduated kernel to reject outliers - corner-specific trajectory error **dropped 80%** on a simulated square-path benchmark, with runtime **50% faster** than batch processing even under 50% outlier noise

Doppler-Based Directional Awareness Wearable, Northeastern University

Sep 2025 - Dec 2025

- A 4-microphone **GCC-PHAT** setup got directional error for a wearable safety device down to **0.13°** in simulation (versus 20-26° at 2-mic spacing), enough to give hearing-impaired pedestrians real spatial awareness of approaching traffic

Experience

Tatum Robotics

Jan 2026 - May 2026

Hardware & Firmware Engineering Co-op

Boston, MA

- Brought up the first Vaydeer USB keypad prototypes on **Linux** (picocom, USB enumeration) before the design moved to RP2040 firmware
- USB dot-input hardware polled at 160ms/dot, so 4-dot chords could take 600ms+ to arrive - fixed it with an **event-based algorithm**: re-press detection, a bitmask prefix check, and a 50ms fallback timer, giving clean chord output at full typing speed
- Designed the Braille Keyboard V1.0 hardware end-to-end in **Altium Designer**, from schematic capture through layout, around an MCP23017 I2C I/O expander and an RP2040 - placed the switch matrix, decoupling caps, and ground-pour cutouts to keyboard-PCB convention, then sent the board to **JLCPCB** for fabrication
- A silent I2C write failure traced back to voltage measurements and an RC time-constant calculation showing the pull-up never reached ACK threshold; the bitbang-I2C workaround held switch reads to **0 errors across 2,509 consecutive polls**
- Ran **seven live sessions across two DeafBlind testers** (five with one, two with another) plus **two official system-test rounds**, mining transcript-level error data to fix two real bugs (mid-word contraction misfires, fast-typing chord merges) and setting Grade 1 as the firmware default based on consistent tester preference

Publication

Patil, R. S., Apraj, A., Bhosale, S., Borikar, S., & Chauhan, A. "Elevating Industrial Picker Arm through Software Mechanic Fusion," IEEE ICNEWS, 2024. ieeexplore.ieee.org/document/10731029